

2021-Ph-H1-Q24

Section: Electricity

Topic: Semiconductors

Which component shows a non-ohmic I–V graph?

Solution:

A **diode** is non-ohmic — current doesn't increase proportionally with voltage and only conducts in one direction.

Answer: B

Guidance for Students:

Ohmic = straight line through origin. Diodes, thermistors, and lamps are non-ohmic.

Revision Tip:

Non-ohmic = not a straight-line graph

Diodes have a sharp “turn-on” voltage